

Weekly Progress Report

Name: Gaurav Kumar

Domain: Data Science And Machine Learning

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Week Ending: 01

I. Overview:

This week focused on understanding the project requirements, exploring the gearbox vibration dataset, extracting meaningful statistical features, and building the initial machine learning pipeline.

II. Achievements:

1. Project Understanding:

- Understood the objective of predictive maintenance using gearbox vibration signals.
- Studied the provided dataset containing Healthy and Broken Tooth gearbox conditions.
- Explored the project documentation and dataset structure.

2. Data Preparation:

Project Name: Gearbox Predictive Maintenance using Machine Learning

- Organized Healthy and Broken Tooth datasets.
- Implemented data loading and preprocessing pipeline.
- Generated structured feature dataset from vibration signals.

3. Feature Engineering:

Implemented statistical feature extraction including:

- Mean
- Standard Deviation
- Minimum
- Maximum

- Peak
- RMS
- Crest Factor

4. Machine Learning Model Development:

Successfully trained multiple machine learning models:

- Decision Tree
- Random Forest
- Logistic Regression
- KNN
- Support Vector Machine
- Gaussian Naive Bayes

III. Challenges:

1. Dataset Understanding:

- Understanding vibration signal data and feature extraction process.
- Selecting useful statistical features.

2. Model Training:

- Comparing multiple algorithms.
- Managing feature consistency across all models.

IV. Learning Resources:

- Scikit-Learn Documentation
- NumPy Documentation
- Pandas Documentation
- Streamlit Documentation
- Machine Learning tutorials

V. Next Week Goals:

- Evaluate all machine learning models.

- Save the best performing model.
- Develop prediction module.
- Build Streamlit web application.

VI. Additional Comments:

“Successfully completed feature engineering and machine learning model training. The project achieved high classification accuracy and is ready for deployment phase.”